

Omid Naeef Nejad

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EDUCATION

- B.Sc. in Computer Science** Sep 2022 – Jul 2026
University of Tehran 
◦ GPA: 17.99/20.00 (Last 2 Years: 18.85/20.00)
◦ Focus: Artificial Intelligence Tehran, Iran

RESEARCH INTERESTS

- Explainable & Unified Multimodal AI (text, image, audio, video)
- Intelligent Multi Agent Systems & Autonomous Decision Making
- Vision Language Models & Object-Centric Scene Understanding
- LLMs as high level decision makers (AVs, robotics, automated systems)

RESEARCH EXPERIENCE

- Vision Researcher (Writing a Survey about Semantic Scene Understanding)** Oct 2025 – Feb 2026
Computer Vision Lab, University of Tehran
Coauthoring an object centric ViT based scene understanding survey (in preparation).
- AI Researcher & Developer** Nov 2024 - Oct 2025
Applied AI Lab, University of Tehran
Contributed to projects in AI for Finance, Digital Transformation, and RAG based chatbot systems.

WORK EXPERIENCE

- AI/BI Engineer (Part-Time)** May 2025 - Nov 2025
Strategy & Development Department, GREEN (Pardis Sanat Siyare Sabz Company) 
◦ Investigated modern data management pipelines and designed business intelligence dashboards. Tehran, Iran
- ZAI Bootcamp Participant** Jan, Feb 2026
Azadi Innovation Factory Tehran, Iran
◦ Gained hands-on experience in business-AI cross-product principles by collaborating with domain expert mentors, focusing on AI product development, MVP creation, and strategic problem-solving.

TEACHING EXPERIENCE

- Chief Teaching Assistant** Feb 2025 - Feb 2026
Managing the TA team, overseeing the development of assignments, and contributing to project design. University of Tehran
◦ Courses: Advanced Information Retrieval, Bio-Computing, Data Structures & Algorithms
- Teaching Assistant** Sep 2024 - Aug 2026
Designed and graded assignments, quizzes, and supported students throughout their studies. University of Tehran
◦ Courses: Data Mining, Basic Programming (twice), Advanced Programming, Modern Information Retrieval, Operating Systems, Compilers

PROJECTS

- Group Activity Recognition for Sports Video Analysis** Bachelor Thesis - Jun 2026
Tools: PyTorch, TimeSformer, Spatio-Temporal Modeling  [Link to GitHub](#)
- Attribute-conditioned Cartoon Face Generation with GAN + diversity aware RL** Dec 2025
Tools: Pytorch, Generative Adversarial Network, Reinforcement Learning  [Link to GitHub](#)
- Mini GPT (Trained on Friends Dialogue Dataset)** Jun 2025
Tools: Pytorch, Transformers  [Link to GitHub](#)
- Modern CNN Architectures & Siamese Networks (Classification and Face Recognition)** Mar 2025
Tools: Pytorch, EfficientNetB0  [Link to GitHub](#)
- Anomaly Detection for Medical Images Using Heterogeneous-AE (Collaborated in a team)** Aug 2025
Tools: Pytorch, Torchvision  [Link to GitHub](#)
- Image Captioning with Encoder-Decoder + Attention** May 2025
Tools: Pytorch, ResNet101, LSTM  [Link to GitHub](#)
- Medical Question Summarization by Round Trip Translation** Jul 2024
Tools: Pytorch, Transformers, NLTK, Scikit-Learn, Googletrans, Scipy  [Link to GitHub](#)
- Speech Emotion Recognition by Mel Spectrograms & HuBERT Embeddings** Aug 2025
Tools: Pytorch, ResNet101, LSTM  [Link to GitHub](#)
- Artificial Intelligence Course Assignments** Oct 2025 - Dec 2025
Tools: Python, L^AT_EX, NumPy  [Link to GitHub](#)
- Data Mining Course Assignments** Feb 2025 - Apr 2025
Tools: Python, Jupyter Notebook, Scikit-Learn, Pandas, Numpy, Matplotlib, LaTeX  [Link to GitHub](#)

• Information Retrieval Course Assignments

Tools: Python, Jupyter Notebook, NLTK, Scikit-Learn, Scipy, Seaborn, Matplotlib

• Metaheuristic Algorithms for NP-Hard Problems (Bio-Computing Course Assignments)

Tools: Python, Jupyter Notebook, Pandas, Numpy, Matplotlib

• C Simplified Compiler

Tools: Java, ANTLR4

• Assembly SQL Parser (Machine Language and Assembly Course Project)

Tools: Assembly

• Morris Mano Basic Computer in Logisim (Principles of Computer Systems Course Project)

Tools: [Logisim]

Oct 2024 - Feb 2025

[Link to GitHub](#)

Feb 2024 - Apr 2024

[Link to GitHub](#)

Apr 2025 - Jun 2025

[Link to GitHub](#)

Jul 2025

[Link to GitHub](#)

Jul 2024

[Link to GitHub](#)

SKILLS

- **Programming:** Python, C++, C, Java, Assembly
- **Data Science, ML & AI:** PyTorch, Transformers, LangChain, TensorFlow, Scikit-Learn, NLTK, Pandas, Matplotlib
- **Databases:** MySQL, relational data modeling, No-SQL, Data Warehouses
- **Tools:** Git/GitHub, Docker (basic), L^AT_EX, Excel
- **Other:** Solid mathematical foundations in linear algebra, probability, optimization

HONORS & COMPETITIONS

• 1st Place — Hackathon Pol 1st

Iran AI Factory, Azadi Innovation Factory

June 2026

[LinkedIn Post](#)

Collaborated in a team to develop a data-driven solution for analyzing and forecasting FMCG sell-in data.

RELEVANT COURSEWORK

- Deep Learning with Applications - Graduate Course (16.75/20)
- Machine Vision - Graduate Course (14.5 / 20)
- Advanced Information Retrieval (20/20)
- Data Mining (20/20)
- Artificial Intelligence (20/20)
- Bio-Computing (19.05/20)
- Non-Linear Programming (17.3/20)
- Design and Analysis of Algorithms (20/20)
- Data Structures and Algorithms (20/20)
- Advanced Programming (17.75/20)
- Basic Programming (20/20)
- Linear Algebra (17/20)
- Probability (17.6/20)
- Principles of Computer Systems (20/20)
- Operating Systems (20/20)

PRESENTATIONS

• Hierarchical Reinforcement Learning

Artificial Intelligence Course, University of Tehran

Dec 2025

[Link to Slide](#)

- Presented the motivation, structure, and key algorithms of hierarchical reinforcement learning, including temporal abstraction, options framework, and benefits over flat RL.

• Web Crawlers and Indexing Pipelines

Advanced Information Retrieval Course, University of Tehran

Dec 2024

[Link to Slide](#)

- Explained the architecture and functioning of web crawlers, indexing pipelines, and critical challenges related to large scale web data processing.

• PCA Theory and Application in Face Recognition

Linear Algebra Course, University of Tehran

Dec 2023

[Link to Slide](#)

- Presented the theoretical foundations of Principal Component Analysis and demonstrated its use in face recognition through eigenfaces, dimensionality reduction, and feature extraction.

VOLUNTEER ACTIVITIES

• AI Instructor – Teaching Team Member

Institute for Research in Fundamental Sciences (IFS) [Globe](#)

Dec 2025 – Present (Biweekly)

- Co-taught AI-focused extracurricular courses in secondary schools, covering Foundations of AI and Prompt Engineering while coordinating lesson design and class management with a co-instructor.

• Seminar Planning Team Member | CS Talks

University of Tehran

Dec 2023 - Jul 2024

- Organized academic seminars, designed promotional content, and improved student engagement.

SELECTED ONLINE CERTIFICATES

- Retrieval Augmented Generation (RAG) with LangChain , DataCamp (2025) [Globe](#) [Certificate](#)
- Deep Reinforcement Learning in Python , DataCamp (2025) [Globe](#) [Certificate](#)
- Reinforcement Learning with Gymnasium , DataCamp (2025) [Globe](#) [Certificate](#)

LANGUAGES

Persian: Native

English: Proficient (TOEFL iBT 89, Reading 25, Listening 21, Speaking 19, Writing 24, December 2025)

Italian: Beginner